

Project 1: Advanced EDA & Feature Engineering

Dataset: Titanic Dataset (891 rows, 12 original columns)
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1. Objective

Transform the raw Titanic dataset into a mathematically clean, ML-ready dataset by handling missing values, neutralizing outliers, and engineering new predictive features.

2. Handling Missing Values

Column	Missing Count	% Missing	Strategy Used	Reasoning
Age	177	~20%	Median Imputation	Age was moderately missing; median avoids distortion from skew, unlike mean.
Embarked	2	~0.2%	Row Deletion (dropna)	Negligible missing count — dropping preserves data integrity without adding synthetic bias.

Column	Missing Count	% Missing	Strategy Used	Reasoning
Cabin	687	~77%	Converted to binary feature (<code>Has_Cabin</code>), original column dropped	Missingness was too extreme (>75%) to impute reliably; presence/absence of a cabin record is itself informative.

```
df['Age'] = df['Age'].fillna(df['Age'].median())
df = df.dropna(subset=['Embarked'])
df['Has_Cabin'] = df['Cabin'].notnull().astype(int)
df = df.drop('Cabin', axis=1)
```

3. Handling Outliers

Column checked: `Fare`

- Mean fare ≈ 32 , but maximum fare was **512.33** — a clear statistical anomaly compared to the 75th percentile (~ 31).

Method used: Interquartile Range (IQR) capping (Winsorization) instead of row deletion, to preserve all 889 rows.

```
Q1 = df['Fare'].quantile(0.25)
Q3 = df['Fare'].quantile(0.75)
IQR = Q3 - Q1
lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR
df['Fare'] = df['Fare'].clip(lower, upper)
```

Result: Maximum fare reduced from 512.33 $\rightarrow$ 65.66, while row count remained unchanged (no data loss).

4. Feature Engineering (3 New Features)

Feature	Formula / Logic	Purpose
FamilySize	<code>SibSp + Parch + 1</code>	Captures total family members aboard, potentially correlated with survival.
IsAlone	<code>1 if FamilySize == 1 else 0</code>	Flags solo travelers, who may have had different survival odds than families.
Title	Extracted from Name (e.g., Mr, Mrs, Miss, Master)	Captures social status/age group not directly present in other columns.

```
df['FamilySize'] = df['SibSp'] + df['Parch'] + 1
df['IsAlone'] = (df['FamilySize'] == 1).astype(int)
df['Title'] = df['Name'].str.extract(r',\s*([^\.,]+)\.')
```

5. Final Dataset

- Rows: 889 (2 dropped due to missing `Embarked`)
 - All missing values resolved
 - Outliers capped, not deleted
 - 3 new engineered features added
 - Saved as `titanic_cleaned.csv`
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6. Key Learnings

- Choice of imputation strategy should depend on the **percentage of missingness**, not applied uniformly.
- Capping (clip) is preferable to deletion when preserving row count/sequence matters.
- Feature engineering from existing columns (like extracting titles from names) can surface predictive signal that raw columns don't expose directly.